

## REPOSITORY 04

# AI, Human Cognition and Knowledge Collapse

Acemoglu, Kong & Ozdaglar (2026)

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[github.com/johnbarraza/ai-04-acemoglu](https://github.com/johnbarraza/ai-04-acemoglu)

# 1. The paper and the agent's problem

## Question

When can accurate, personalized AI improve decisions today yet erode the shared knowledge needed tomorrow?

## One mechanism

Human effort jointly produces a private signal and a public externality. AI adds private precision only.

Choice:  $e \geq 0$

$$\max_{e \geq 0} \underbrace{f_{00} + G(X)\Delta_G + G(X)G(Y)\Delta_X}_{\text{expected task output}} - \frac{\varepsilon}{\varepsilon + 1} e^{(\varepsilon+1)/\varepsilon},$$

$$Y = \sigma^{-2} + \lambda_I e + \tau_A, \quad \Delta_I = 0, \quad \Delta_X > 0.$$

For  $X > 0$ , the unique interior optimum satisfies

$$\Delta_X G(X) \lambda_I g(Y) = e^{1/\varepsilon}.$$

Source: May 5, 2026 PDF, Sections 3.1–3.5, equations (4)–(6).

## 2. Main result: one complement, one substitute

**Observation 1 (corrected domain).** Suppose Gaussian posterior errors,  $X > 0$ ,  $Y > 0$ ,  $\lambda_I > 0$ ,  $\tau_A \geq 0$ , and Assumption 1 ( $\Delta_I = 0$ ,  $\Delta_X > 0$ ). With  $G(z) = 2\Phi(\sqrt{z}) - 1$  and  $g = G'$ , then

$$U_{eX} = \Delta_X \lambda_I g(X) g(Y) > 0$$

$$U_{e\tau_A} = \Delta_X G(X) \lambda_I g'(Y) < 0.$$

### General knowledge $X$

Raises the payoff from learning the context: effort and public precision are complements.

### Agentic precision $\tau_A$

Enters the same precision index as private learning: AI and effort are substitutes.

**Implication.** The unique best response  $e(X, \tau_A)$  rises in  $X$  and falls in  $\tau_A$ , strictly for  $X > 0$ .

Source: Observation 1 and Observation 2, pp. 13–15. The paper omits  $X > 0$  from the displayed cross-partial domain.

### 3. What I did: attack the zero-collapse hinge

I restored the production term the paper never relaxes:

$$\Delta_I > 0 \implies U_e = \lambda_I g(Y) \{ \Delta_I + G(X) \Delta_X \} - e^{1/\varepsilon}.$$

At the alleged collapse boundary,

$$U_e(0; 0, \tau_A) = \lambda_I g(\sigma^{-2} + \tau_A) \Delta_I > 0.$$

Hence  $e(0, \tau_A) > 0$  and

$$F(0) = [\Sigma^2 + (\lambda_G l e(0, \tau_A))^{-1}]^{-1} > 0.$$

Source: Original extension; baseline Assumption 1 and Sections 5.1–5.3.

#### Economic consequence

Exact zero knowledge is no longer a fixed point. A low-knowledge trap may survive, but literal collapse does not.

#### Formal check

EconCSLib target:

AK026KnowledgeCollapse. Lean checks the algebraic/order core; source fidelity remains a separate audit.

## 4. Where I did not believe the AI

Add your own  
hand derivation

hand/derivation.jpg

Source: Own derivation and audit against the May 5, 2026 PDF.

Acemoglu–Kong–Ozdaglar (2026) · model audit

### Verdict

The sign calculation is correct on  $X > 0$ , but it does not identify something unique to an *agent*: any context-specific precision shock added to  $Y$  crowds out effort.

The stronger claim—complete destruction of useful knowledge—leans on  $\Delta_l = 0$ . It is better read as a sharp benchmark for a low-knowledge trap. **Two slips to**

**remember:**

- ▶  $g(X)$  is not finite at the endpoint  $X = 0$ ;
- ▶ Proposition 5 leaves  $\tau_A = \tau_A^c$  unstated.